

# Cleaning Messy Data

## Task 3 – Cleaning Data

### Oasis Infobyte

Submitted by: Bhavyaa Bansal

Track: Data Analytics

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#### Introduction

Data cleaning is an essential step in the data analysis workflow. Raw datasets often contain missing values, duplicate records, inconsistent formatting, incorrect data types, and invalid observations that can negatively impact analytical results. The objective of this project was to clean a deliberately messy Google Play Store dataset and transform it into a reliable, analysis-ready dataset by applying systematic data preprocessing techniques.

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#### Dataset Description

The project utilized the Google Play Store dataset containing information about Android applications. The dataset includes attributes such as application name, category, rating, number of reviews, application size, installation count, pricing information, content rating, genres, last update date, application version, and supported Android version.

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#### Data Quality Assessment

The initial assessment involved evaluating the overall quality of the dataset by identifying missing values, duplicate records, incorrect data types, and potential anomalies.

The following checks were performed:

- Dataset dimensions
- Missing value analysis
- Duplicate row detection
- Data type inspection
- Descriptive statistical summary
- Validation of numeric ranges

This assessment provided a clear understanding of the issues that required cleaning.

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#### Data Cleaning Process

##### Missing Value Handling

Different imputation techniques were selected according to the characteristics of each variable.

- Missing values in the **Rating** column were replaced using the **median**, as ratings are numerical and the median is less sensitive to outliers.
  - Missing values in **Type**, **Content Rating**, and **Android Version** were filled using the **mode**, preserving the most frequently occurring category.
  - Missing values in **Current Version** were replaced with "**Unknown**" because version information could not be reliably inferred.
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## Duplicate Removal

Duplicate records were identified using Pandas' duplicate detection methods and removed to prevent redundant observations from influencing future analysis.

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## Standardization

Several formatting inconsistencies were corrected to improve data consistency.

The following standardization tasks were completed:

- Converted the **Last Updated** column into datetime format.
  - Standardized installation counts by removing commas and "+" symbols before converting them into integers.
  - Standardized application size values into a consistent numerical format.
  - Corrected price formatting where necessary.
  - Removed unnecessary whitespace from text columns.
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## Data Type Correction

Appropriate data types were assigned to each feature.

Examples include:

- Ratings and prices as floating-point values.
- Reviews and installs as integers.
- Last updated dates as datetime values.

These corrections improve computational efficiency and ensure compatibility with analytical methods.

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## Outlier Detection

Outliers were identified using the **Interquartile Range (IQR)** method across numerical variables including ratings, reviews, installs, application size, and price.

Although statistical outliers were detected, most were retained because they represented legitimate applications with exceptionally high popularity or premium pricing rather than erroneous observations. Only clearly invalid records identified during the cleaning process were removed.

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## Results

The completed cleaning process significantly improved the overall quality of the dataset.

Key improvements include:

- Missing values appropriately handled.
- Duplicate records removed.
- Invalid observations eliminated.
- Consistent formatting achieved across multiple columns.
- Correct data types assigned.
- Clean dataset successfully exported for future analysis.

The cleaned dataset is now suitable for exploratory data analysis, visualization, and predictive modelling.

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## **Conclusion**

This project demonstrated a complete end-to-end data cleaning workflow using Python and Pandas. By applying appropriate preprocessing techniques, the dataset was transformed into a consistent, reliable, and analysis-ready format. Proper documentation of each cleaning decision ensures transparency, reproducibility, and adherence to professional data analytics practices, making the dataset suitable for subsequent analytical and machine learning tasks.